Mehery PushApp Integration Guide (App Brew / Natori)

August 11, 2026

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Mehery PushApp Integration Guide (App Brew / Natori)

Audience: Client engineering team integrating or maintaining PushApp in this host app.
Branch: `pushapp_neilo`
Host app: App Brew Natori (`com.natori.appbrew`)
SDK (read-only): `react-native-mehery-event-sender` from local path `../msdk/react-native-mehery-event-sender`
Date: 2026-08-10

This document explains **what we changed**, **where**, **why**, and **how to repeat or extend** the integration. The Mehery SDK source itself was **not** modified.

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1. Purpose and architecture

PushApp (Mehery) provides:

- Device registration and authenticated API access (`pushappAuth` + `initSdk`)
- Rich push (Android native FCM service; iOS NSE + NCE carousel / action categories)
- In-app poll overlays (`PollOverlayProvider`)
- Event / journey tracking (`sendCustomEvent`, login link, page lifecycle)

App Brew already had Firebase Analytics / `@gauntlet/firebase-push` / Notifee. This integration:

1. Boots Mehery on cold start.
2. **Dual-sends** every Excel/GA4 commerce event to Mehery via a new `MeheryTracker` on `AnalyticsProvider` (no patches to `@gauntlet/*` call sites).
3. Lets **Mehery own PushApp rich tray UI** while keeping `modules['push'] = FirebasePush.getInstance` for the AppBrew push module contract.

```
User action → AnalyticsProvider
                ├─ FirebaseTracker / AppbrewTracker / FacebookTracker  (release only, !__DEV__)
                └─ MeheryTracker → sendCustomEvent (always, including __DEV__)

FCM data push → Android MyFirebaseMessagingService (Mehery)
            → iOS ImageNotification NSE + ImagePreviewExtension NCE
```

---

2. Prerequisites checklist

Before shipping:

- ☐ Firebase configs present (`android/app/google-services.json`, `ios/GoogleService-Info.plist`) — currently sample Natori project; replace for production
- ☐ Google Services Gradle plugin applied (already present)
- ☐ Peer packages installed: `@react-native-firebase/app`, `@react-native-firebase/messaging`, `@react-native-async-storage/async-storage`, `react-native-push-notification`
- ☐ Real PushApp credentials in Info.plist + `strings.xml` (replace `pa_REPLACE_*` / `pas_REPLACE_*`)
- ☐ Real channel id in `src/mehery/bootstrap.ts` (`MEHERY_CHANNEL_ID`)
- ☐ Android: RNFB default messaging service removed (`tools:node="remove"`) — **done**
- ☐ iOS: Push capability + `UIBackgroundModes` → `remote-notification` — **done**
- ☐ iOS: Notification content extension target signed (ImagePreviewExtension)
- ☐ Cold start: credentials → `pushappAuth` → `initSdk` before events — **wired**
- ☐ `PollOverlayProvider` mounted — **done**

---

3. Package install

DEPENDENCY LINES ADDED TO `PACKAGE.JSON`

```
"react-native-mehery-event-sender": "file:../msdk/react-native-mehery-event-sender",
"react-native-push-notification": "^8.1.1",
"@react-native-async-storage/async-storage": "^3.1.1"
```

Existing (kept): `@react-native-firebase/app` and `@react-native-firebase/messaging` at `23.3.1`.

COMMANDS

```
cd "/Users/neil/Desktop/APP Brew"
pnpm add react-native-mehery-event-sender@file:../msdk/react-native-mehery-
event-sender \
  react-native-push-notification \
  @react-native-async-storage/async-storage

pnpm pod-install   # ios: bundle install && bundle exec pod install
```

CocoaPods includes `MeheryEventSender (0.0.49)` after install.

---

4. Android native edits

4.1 `ANDROID/APP/SRC/MAIN/ANDROIDMANIFEST.XML`

Added:

- `WAKE_LOCK` permission
- Service removal so Mehery's `MyFirebaseMessagingService` owns FCM:

```
<service

android:name="io.invertase.firebase.messaging.ReactNativeFirebaseMessagingService"
  android:exported="false"
  tools:node="remove" />
```

Why: React Native Firebase registers a no-op messaging service. If both remain, background rich push (images, carousel, CTAs) can hit the wrong service.

Already present: `INTERNET`, `POST_NOTIFICATIONS`, Firebase default channel / icon meta-data. Mehery SDK merges its own service/receivers from the library manifest.

4.2 `ANDROID/APP/SRC/MAIN/RES/VALUES/STRINGS.XML`

Added credential placeholders:

String resource	Purpose
`mehery_prod_app_id` / `mehery_prod_app_key`	Production (`initSdk` env `false`)
`mehery_sandbox_app_id` / `mehery_sandbox_app_key`	Sandbox (`true`)
`mehery_dev_app_id` / `mehery_dev_app_key`	Development (`'development'`)

Replace `pa_REPLACE_*` / `pas_REPLACE_*` with client dashboard values. Rotating credentials requires a store update.

4.3 UNCHANGED BUT REQUIRED

- `android/build.gradle` — `com.google.gms:google-services`
- `android/app/build.gradle` — `apply plugin: "com.google.gms.google-services"`
- `android/app/google-services.json`

4.4 ANDROID NOTIFICATION TYPES (PROVISIONED BY MEHERY NATIVE PATH)

Type	How it triggers
Simple banner	title/body, no multi-image
BigPicture	single `image` / `imageUrl` / `image_url`
Carousel	`imageUrls` / `image_urls` / `carousel_images` / `image1`… (max 4); Prev/Next
Live tray	`message1` + `message2` + `message3` (+ progress / hero)
CTA buttons	`cta_buttons` or `titleN` / `urlN` / `actionN`
Deep link	`notification_url` / `notificationUrl`
Local test	JS `triggerCarouselNotification({ title, body, images })` (Android-only)

Payload rule: Background rich UI requires **data-only**, high-priority FCM. A top-level FCM `notification` block shows the stock OS tray and skips Mehery's service.

Verify:

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```
adb logcat -s MyFirebaseMessagingService:I
```

Example carousel `data` map:

```
{
  "title": "New arrivals",
  "body": "Swipe looks",
  "imageUrls": "[\"https://example.com/1.jpg\",\"https://example.com/2.jpg\"]",
  "notification_url": "https://natori-appbrew.myshopify.com/collections/all",
  "title1": "Shop",
  "url1": "https://natori-appbrew.myshopify.com/collections/all",
  "action1": "SHOP"
}
```

---

5. iOS native edits

5.1 `IOS/NATORI/INFO.PLIST`

Added:

- `MeheryProdAppId` / `MeheryProdAppKey`
- `MeherySandboxAppId` / `MeherySandboxAppKey`
- `MeheryDevAppId` / `MeheryDevAppKey`
- `UIBackgroundModes`: `remote-notification`, `fetch`
- `NSSupportsLiveActivities` (+ frequent updates) for Live Activity

Entitlements already include `aps-environment`.

5.2 `IOS/NATORI/APPDELEGATE.SWIFT`

Added:

- `registerNotificationCategories()` — all PushApp 2-button and 3-button categories (including `CAROUSEL_CATEGORY`, `THREE_CAROUSEL_CATEGORY`, …)
- `userNotificationCenter(_:willPresent:)` — show banner/list/sound/badge in foreground
- `userNotificationCenter(_:didReceive:)` — open `notification_url` / related keys via `urlForNotificationBody`

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5.3 NOTIFICATION SERVICE EXTENSION (`IOS/IMAGENOTIFICATION/`)

Replaced Firebase `Messaging.serviceExtension().populateNotificationContent` with Mehery's multi-image NSE (`NotificationService.swift` from the SDK example).

Updated `Info.plist` with `NSAppTransportSecurity` → `NSAllowsArbitraryLoads` so the extension can download media.

Podfile: ImageNotification target no longer depends on Firebase Messaging (Mehery NSE is pure UserNotifications).

5.4 NOTIFICATION CONTENT EXTENSION (`IOS/IMAGEPREVIEWEXTENSION/`)

Copied from Mehery example (carousel / category UI). Categories declared in extension `Info.plist` match AppDelegate.

Xcode: `ImagePreviewExtension` native target added (`com.natori.appbrew.ImagePreviewExtension`), embedded in the app. **You must** create a matching provisioning profile / App Store Connect capability for the new bundle id before release builds.

5.5 LIVE ACTIVITY (`IOS/DELIVERYACTIVITY/`)

Status: wired and compiling (simulator build of `DeliveryActivityExtension` succeeded).

Item	Value
Xcode target	`DeliveryActivityExtension` (embedded in `natori`)
Bundle ID	`com.natori.appbrew.DeliveryActivity`
App Group	`group.com.natori.appbrew.NotificationLiveActivity` (main + extension)
Start/update	`ios/natori/MeheryLiveActivity.swift` via `didReceiveRemoteNotification`
UI	`DeliveryActivityLiveActivity.swift` (Lock Screen / Dynamic Island)

Client must still: create the App Group + App IDs/profiles in Apple Developer and assign signing for `DeliveryActivityExtension`.

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Trigger: background/content-available push with `activity_id` plus live template fields (`message1` – `message3`, `live_activity`, progress, or image URL). See `ios/DeliveryActivity/README-NATORI.md`.

Example app sources `DeliveryActivity.swift` / ControlWidget were left on disk for reference but are **not** compiled (iOS 18 ControlWidget / AppIntent home-widget sample).

5.6 PODS

```
pnpm pod-install
```

Confirms `MeheryEventSender (0.0.49)` and extension targets integrate.

---

6. JavaScript / TypeScript wiring

6.1 `INDEX.JS`

```
import { registerFcmBackgroundHandler } from 'react-native-mehery-event-sender';
registerFcmBackgroundHandler();
// then AppRegistry.registerComponent(...)
```

Must run **before** `AppRegistry.registerComponent`.

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6.2 `src/mehery/` MODULE FOLDER

File	Role
`bootstrap.ts`	Credentials → `pushappAuth` → `initSdk` → `waitForSdkReady` → deep links → Android permission → `OnAppLaunch` / `OnAppOpen`
`MeheryTracker.ts`	Dual-send GA4 events + identity via `setUserDetails`
`identity.ts`	Thin wrappers for login / logout / profile
`navigation.ts`	`OnPageOpen` / `OnPageClose` helpers
`deepLinks.ts`	`setNotificationUrlHandler` + host rewrites for Shopify hosts
`pushCoexistence.ts`	Android permission + PushApp payload heuristic
`index.ts`	Public exports

Defaults (change before prod):

- `MEHERY_ENVIRONMENT = 'development'` → host `pushapp.co.in`
- `MEHERY_CHANNEL_ID = 'natori_REPLACE_CHANNEL_ID'`

6.3 `src/app/app.tsx`

- Registers `new MeheryTracker()` on `AnalyticsProvider` **inside and outside** `__DEV__` (Firebase/Appbrew/Facebook stay `!__DEV__` only).
- Keeps `modules['push'] = FirebasePush.getInstance`.
- `useEffect` calls `bootstrapMeherySdk()`.
- Mounts `<PollOverlayProvider />` beside `Shell`.

6.4 OPTIONAL IN-APP POLL SURFACES

`PollOverlayProvider` is required for overlay polls. Optionally place:

- `InlinePollContainer`
- `TooltipPollContainer`

Poll styles supported by the SDK: roadblock, floater, banner, picture-in-picture, bottomsheet, inline, tooltip.

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7. Credential and environment matrix

`initSdk` env	Host	iOS keys	Android strings
`false`	`pushapp.ai`	`MeheryProdAppId` / `MeheryProdAppKey`	`mehery_prod_*`
`true`	`pushapp.net.in`	`MeherySandboxAppId` / `MeherySandboxAppKey`	`mehery_sandbox_*`
`'development'`	`pushapp.co.in`	`MeheryDevAppId` / `MeheryDevAppKey`	`mehery_dev_*`

Always call `readCredentialsForEnvironment(environment)` then `pushappAuth` with the **same** environment you pass to `initSdk`.

---

8. Bootstrap sequence (mandatory)

On every cold start:

1. `readCredentialsForEnvironment(env)`
2. `pushappAuth(xApiId, xApiKey)`
3. `await initSdk(null, '<tenant>_<channel>', env, logs)`
4. `await waitForSdkReady()`
5. Mount / keep `PollOverlayProvider`
6. Then fire identity / page / custom events

Implemented in `src/mehery/bootstrap.ts` and invoked from `App.tsx`.

---

9. Dual-send: GA4 / Excel events → Mehery

Mechanism: `MeheryTracker` extends `AnalyticsTrackerV2` and implements `sendEvent` → `sendCustomEvent(exactGa4Name, payload, { eventType: 'INTERACTIVE' })`.

`app_install_ios` / `app_install_android` are remapped to GA4 name `app_install`.

`setUserDetails` → `OnUserLogin` + `updateUserProfile`; clearing user → `OnUserLogOut`.

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Interactive events fired before device link are **queued** by the SDK (cap 50) and flushed after successful login.

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FULL EVENT CATALOG (FROM `APP-BREW EVENTS WITH`

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`PROPERTIES.XLSX`)

Event	Properties dual-sent
`app_install`	`version`, `sizeWidth`, `sizeHeight`
`screen_view`	`screen_name`, `screen_class`
`push_notification_subscribed`	`push_enabled`
`push_notification_unsubscribed`	`push_enabled`
`login`	`method`
`signup`	`method`
`logout`	*(none)*
`view_item`	`currency`, `value`, `items[]` (`item_id`, `handle`, `item_name`, `item_variant`, `price`, `original_price`, `discount`, `url`, `sku`, `item_category`, `item_brand`, `img_url`)
`view_item_list`	`item_list_id`, `handle`, `item_list_name`, `url`
`search`	`search_term`
`add_to_cart`	`currency`, `value`, `items[]` (incl. `quantity`, `sku`, …)
`remove_from_cart`	`currency`, `value`, `items[]`
`view_cart`	`url`, `value`, `checkout_token`, `sub_total`, `discount`, `currency`, `coupon`, `items[]`
`add_to_wishlist`	`currency`, `value`, `items[]`
`remove_from_wishlist`	`currency`, `value`, `items[]`
`notify_back_in_stock`	`currency`, `value`, `items[]`
`begin_checkout`	cart-like set
`add_shipping_info`	cart-like set
`apply_coupon`	`coupon`, `isApplicable`
`remove_coupon`	`coupon`

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Event	Properties dual-sent
`purchase`	`transaction_id`, `shopify_order_id`, `url`, `value`, `checkout_token`, `sub_total`, `discount`, `currency`, `coupon`, `items[]`
`refund`	`reason`, `transaction_id`, `value`, `currency`

Lifecycle outside the analytics bus (after init): `OnAppLaunch`, `OnAppOpen`.

---

10. Notification matrices

ANDROID

Kind	Notes
Simple	title/body
BigPicture	single image key
Carousel	≥2 images; Prev/Next; at most 1 CTA (action slots)
Live tray	`message1` – `message3`
CTAs	`cta_buttons` or indexed title/url/action
Deep link	`notification_url`
Click track	`t` / `click_token` → `/v1/notification/push/track`
Local test	`triggerCarouselNotification`

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IOS

Kind	Notes
Plain banner	APNs alert
Single-image rich	NSE attachment
Carousel	NSE multi-image + NCE (`CAROUSEL_CATEGORY` / `THREE_CAROUSEL_CATEGORY`)
Action categories	Full 2-button and 3-button sets registered in AppDelegate
Silent daily ping	payload `type: silent_daily_ping`
Body URL	`notification_url` via AppDelegate / JS handler
Live Activity	Wired (`DeliveryActivityExtension` + App Group); Apple Developer signing required

---

11. Coexistence with FirebasePush / Notifee

Layer	Owner
AppBrew `modules['push']` API / permissions contract	`@gauntlet/firebase-push`
PushApp rich tray (background Android)	Mehery `MyFirebaseMessagingService`
PushApp rich tray (iOS)	Mehery NSE + NCE
Dual-send analytics	`MeheryTracker`
GA4 / Appbrew / Facebook analytics	Existing trackers (`!__DEV__`)

Duplicate tray: When a PushApp data payload arrives in the foreground, Mehery's JS path may display it. If Notifee/FirebasePush also displays the same FCM message, use `isLikelyPushAppPayload(data)` from `src/mehery/pushCoexistence.ts` in any custom foreground handler to skip a second local notification. Do **not** modify the Mehery SDK to reconcile conflicts.

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12. E2E verification checklist

BUILD / SDK / IDENTITY

Check	Status
`pnpm install` with local Mehery package	Done
`pnpm pod-install` includes `MeheryEventSender`	Done
Cold-start bootstrap wired	Done (device run needed to confirm no throw)
`PollOverlayProvider` mounted	Done
Android manifest RNFB remove	Done
Login / logout dual path via tracker	Wired; device login needed
Real credentials / channel id	**Blocked** — placeholders pending client

DUAL-SEND (22 EXCEL EVENTS)

Check	Status
`MeheryTracker` whitelists all catalog events	Done
Registered in `__DEV__` and production	Done
Device: each event visible in PushApp / SDK logs	Pending device QA

ANDROID NOTIFICATIONS

Check	Status
Native service path provisioned via SDK	Done (library merge)
Data-only high-priority sends	Pending PushApp / FCM console
`triggerCarouselNotification` smoke	Pending device
logcat `MyFirebaseMessagingService`	Pending device

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IOS NOTIFICATIONS

Check	Status
Categories registered	Done
NSE Mehery multi-image	Done (source swapped)
NCE target present	Done (signing profile required)
Carousel / CTA device test	Pending device
Live Activity	Target builds; device QA after App Group + signing

COEXISTENCE

Check	Status
FirebasePush module kept	Done
Double-tray guard helper	Done (`isLikelyPushAppPayload`)
Deep link rewrites for Shopify hosts	Done

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13. File change inventory

Path	Change
`package.json` / `pnpm-lock.yaml`	Mehery + peers
`index.js`	`registerFcmBackgroundHandler`
`src/app/App.tsx`	Bootstrap, tracker, `PollOverlayProvider`
`src/mehery/*`	New integration helpers
`android/.../AndroidManifest.xml`	WAKE_LOCK + RNFB service remove
`android/.../values/strings.xml`	Mehery credentials
`ios/natori/Info.plist`	Credentials + background modes + Live Activity flags
`ios/natori/AppDelegate.swift`	Categories + presentation + URL open + Live Activity background hook
`ios/natori/MeheryLiveActivity.swift`	Start/update Live Activities from push
`ios/ImageNotification/*`	Mehery NSE
`ios/ImagePreviewExtension/*`	Mehery NCE sources + Xcode target
`ios/DeliveryActivity/*`	Live Activity widget extension (compiles; App Group signing pending)
`ios/natori/natori.entitlements`	App Group for Live Activity images
`ios/Podfile`	Extension targets; drop Firebase from NSE
`ios/Podfile.lock` / Pods	MeheryEventSender
`.cursor/skills/mehery-pushapp-integration/*`	Agent skill
`docs/mehery-pushapp-integration-guide.md` (+ pdf/docx)	This handoff

Never edited: `Desktop/msdk/react-native-mehery-event-sender/**`

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14. Troubleshooting

Symptom	Likely cause	Fix
Auth / init fails	Placeholder credentials or wrong env	Match env + native keys + channel id
No rich Android background push	Top-level FCM `notification` or dual messaging services	Data-only + confirm RNFB remove; check logcat
Carousel missing on iOS	NCE not embedded / not signed / wrong category	Confirm ImagePreviewExtension + category ids
Events missing in PushApp	Fired before `initSdk` / before login queue flush	Await bootstrap; login then retry
Double foreground notification	Notifee + Mehery both display	Skip Notifee when `isLikelyPushAppPayload`
Live Activity not showing	App Group / signing / payload missing `activity_id`	Enable `group.com.natori.appbrew.NotificationLiveActivity`; sign `DeliveryActivityExtension`; see `README-NATORI.md`

---

15. What we did not change (and why)

The Mehery SDK under `Desktop/msdk/react-native-mehery-event-sender` is treated as a **vendor package**. All host-specific behavior lives in this App Brew repo so upgrades of the SDK remain clean.

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16. Appendix — public API summary

Auth/init: `readCredentialsForEnvironment`, `pushappAuth`, `initSdk`, `waitForSdkReady`
Identity: `OnUserLogin`, `OnUserLogOut`, `updateUserProfile`
Events: `sendCustomEvent`, `flushPendingCustomEvents`, `OnPageOpen`, `OnPageClose`, `OnAppOpen`, `OnAppLaunch`
Push helpers: `registerFcmBackgroundHandler`, `ensureAndroidNotificationPermission`, `triggerCarouselNotification`, `setNotificationUrlHandler`, `configureNotificationLinkRewrites`, `openNotificationLink`
UI: `PollOverlayProvider`, `InlinePollContainer`, `TooltipPollContainer`

FURTHER READING (SDK, READ-ONLY)

- `../msdk/react-native-mehery-event-sender/README.md`
- `../msdk/react-native-mehery-event-sender/example/src/App.tsx`
- `../msdk/react-native-mehery-event-sender/example/ios/` (NSE / NCE / DeliveryActivity / AppDelegate categories)

AGENT SKILL (INTERNAL)

`.cursor/skills/mehery-pushapp-integration/SKILL.md` and `reference.md`

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